ض BOG'CHA GUARD AI

Aqlli Bog'cha Monitoring Tizimi

Project Requirements Document (PRD / TZ)

Versiya: v1.0 (MVP)	**Sana:** 2025
Status: Draft	**Muallif:** Solution Architect Team

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1. Loyiha Haqida Umumiy Ma'lumot

1.1 Muammo va Kontekst

Zamonaviy bog'chalarda bolalar xavfsizligini ta'minlash tobora muhim masalaga aylanmoqda. Tarbiyachilar bir vaqtning o'zida ko'p bolani nazorat qilishda qiyinchilikka duch keladi, xavfli holatlar vaqtida aniqlanmaydi va ota-onalar farzandlarining bog'chadagi hayoti haqida to'liq ma'lumotga ega emas.

Muammo statistikasi (global)

- Bolalar bog'chalarda har yili 30%+ jarohatlari nazorat etilmagan vaqt davomida sodir bo'ladi
- Tarbiyachilar o'rtacha 1 kishiga 8-12 bola to'g'ri keladi — real vaqtda monitoring qiyin
- Bog'cha hududidan bolaning chiqib ketishi hodisalari yiliga 100,000+ global
- Begona shaxslarning bog'chaga kirishi 40% hollarda xodimlar tomonidan ushlanmaydi

1.2 Yechim: BogchaGuard AI

BogchaGuard AI — bu kompyuter ko'rish (Computer Vision) va sun'iy intellekt asosida qurilgan real-vaqt monitoring tizimi. Kameralar orqali bog'cha ichida va hududida bo'lgan barcha holatlar avtomatik tahlil qilinadi va tegishli xodimlarga zudlik bilan xabardorlik yuboriladi.

Asosiy afzalliklar

Real-vaqt AI tahlili — hodisa sodir bo'lganda darhol aniqlash

3 darajali rol tizimi — har bir xodim faqat o'z doirasidagi ma'lumotni ko'radi

Push-notification tizimi — mobil va web orqali zudlik bilan xabardorlik

Markazlashgan dashboard — mudirlar uchun butun bog'cha ko'rinishi

Kengaytiriladigan arxitektura — yangi kameralar va filiallarni oson qo'shish

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2. Foydalanuvchilar va Rollar

2.1 Rol Matritsasi

Tizim uchta asosiy rol asosida qurilgan. Har bir rol o'z vakolat doirasi va ma'lumot ko'rinishiga ega:

ROL	JAVOBGARLIK	NOTIFICATION TURLARI
¢ Tarbiyachi	O'z guruhidagi bolalarni nazorat qilish	Yiqilish, mushtlashuv, xavfli o'yin, bola yig'lashi, favqulodda holat
Ą Xavfsizlik	Bog'cha hududini nazorat qilish	Bola hududdan chiqib ketishi, begona shaxs kirishi, ruxsatsiz kirish
´ Mudir	Barcha jarayonlarni nazorat qilish	Barcha yuqoridagi + xodimlar harakati, soatlik hisobotlar, shubhali faoliyat

2.2 Tarbiyachi Roli — Batafsil

Tarbiyachi ko'rishi mumkin bo'lgan ma'lumotlar

- Faqat o'z guruhiga biriktirilgan kameralar streamini ko'radi
- O'z guruhidagi bolalar ro'yxati va ularning holati (present/absent)
- Guruhiga oid hodisalar tarixi va arxivi
- Guruh statistikasi (bugungi hodisalar soni, davomatlik)

Tarbiyachiga keladigan alertlar

‘ KRITIK: Bola yiqildi / jarohatlandi
‘ KRITIK: Bolalar o'rtasida jismoniy to'qnashuv
È OGOHLANTIRISH: Uzoq vaqt yig'lash (>3 daqiqa)
È OGOHLANTIRISH: Xavfli joyga yaqinlashish (derazalar, katta mebel)
’ AXBOROT: Guruh bola soni o'zgarishi

2.3 Xavfsizlik Xodimi Roli — Batafsil

Xavfsizlik xodimiga keladigan alertlar

‘ KRITIK: Bola bog'cha hududidan tashqariga chiqdi
‘ KRITIK: Biometrikada tanilmagan shaxs kirishga harakat qildi
‘ KRITIK: Kirish nuqtasida shubhali faoliyat
È OGOHLANTIRISH: Tashqi hududda tanilmagan transport vositasi
È OGOHLANTIRISH: Yopiq soatlarda harakat aniqlandi

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2.4 Mudir Roli — Batafsil

Mudir ko'rishi mumkin bo'lgan to'liq ma'lumotlar

- Barcha guruhlar va xonalarning real-vaqt kamera streami
- Barcha tarbiyachilar va xavfsizlik xodimlarining harakati (login/logout vaqtlari)
- Bog'cha bo'yicha kunlik/haftalik/oylik hisobotlar
- Barcha hodisalarning to'liq arxivi va tahlili
- Xodimlar samaradorligi va javob vaqti statistikasi
- Ota-onalar bilan muloqot tarixi

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3. Funksional Talablar

3.1 Camera & Computer Vision Moduli

3.1.1 Kamera Integratsiyasi

Tizim quyidagi kamera protokollarini qo'llab-quvvatlaydi:

- RTSP (Real Time Streaming Protocol) — IP kameralar uchun asosiy protokol
- ONVIF standarti — ko'plab ishlab chiqaruvchilarning kameralari bilan muvofiqlik
- HTTP/MJPEG stream — arzonroq kameralar uchun fallback
- WebRTC — web browserda past kechikish bilan ko'rsatish uchun

3.1.2 AI Detection Qobiliyatlari

Computer Vision moduli quyidagi hodisalarni real vaqtda aniqlay olishi shart:

HODISA TURI	TEXNOLOGIYA	ANIQLIK DARAJASI (MIN)
Odam tanish (Face Recognition)	FaceNet / DeepFace	>95%
Yiqilishni aniqlash (Fall Detection)	Pose Estimation (MediaPipe)	>90%
Mushtlashuvni aniqlash (Fight Detection)	Action Recognition (SlowFast)	>85%
Yig'lashni aniqlash (Cry Detection)	Audio + Video fusion	>88%
Hudud chegarasini kuzatish (Zone Monitoring)	Object Tracking (YOLO+ByteTrack)	>97%
Noma'lum shaxsni aniqlash (Intruder Detection)	FaceNet + Cosine Similarity	>92%
Xavfli zona yaqinlashuvi	Semantic Segmentation	>93%
Harakat tahlili (Activity Recognition)	Optical Flow + CNN	>82%

3.1.3 Video Processing Pipeline

Har bir kamera uchun quyidagi processing pipeline qo'llaniladi:

Video Processing Zanjiri

1. Camera Stream → RTSP Ingestor (FFmpeg) → Frame Extraction (15 FPS)
2. Frame → Pre-processing (resize 640x480, normalize) → AI Inference Engine
3. AI Output → Event Classifier → Alert Generator → Notification Service
4. Parallel: Frame → Video Encoder → Storage Service (MP4 H.264)

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5. Storage: Last 30 kunlik video arxivi (avtomatik eski fayllarni o'chirish)

3.2 Notification (Alert) Tizimi

3.2.1 Alert Darajalari

DARAJA	RANG	JAVOB VAQTI
KRITIK (P0)	‘ Qizil	Zudlik bilan — 0-5 soniya ichida
OGOHLANTIRISH (P1)	È Sariq	Tez — 5-30 soniya ichida
AXBOROT (P2)	’ Ko'k	Normal — 30 soniya - 2 daqiqa

3.2.2 Notification Kanallari

- In-app notification — web dashboard va mobil ilova (real-vaqt WebSocket)
- Push notification — Firebase Cloud Messaging (FCM) — mobil qurilmalar
- SMS — Eskiz.uz yoki Playmobile API (O'zbekiston SMS gateway)
- Email — muhim hodisalar uchun (mudir va ma'muriyat)

3.3 DATA Tizimi

3.3.1 Bolalar Ma'lumotlar Bazasi

Har bir bola uchun tizimda quyidagi ma'lumotlar saqlanadi:

MA'LUMOT	TAVSIF
Ism, familiya, tug'ilgan sana	Asosiy identifikatsiya
Guruh biriktirilishi	Qaysi guruhga tegishli
Yuz rasmi (biometrik)	Face Recognition uchun fotosuratlar (3-5 ta)
Ota-ona kontaktlari	Xabardorlik yuborish uchun
Tibbiy ma'lumotlar	Allergiya, kasalliklar (maxfiy)
Davomatlik tarixi	Kelgan/ketgan vaqtlari
Hodisalar tarixi	Ushbu bola bilan bog'liq barcha eventlar

3.3.2 Xodimlar Ma'lumotlar Bazasi

MA'LUMOT	TAVSIF
Ism, familiya, lavozim	Asosiy identifikatsiya
Guruh biriktirilishi	Tarbiyachilar uchun

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Login/Logout vaqtlari	Mudir kuzatuvi uchun
Yuz rasmi (biometrik)	Tizimga kirish va kuzatuv uchun
Ruxsatlar va rollar	RBAC — Role-Based Access Control
Faoliyat logi	Audit trail — barcha amallar yozib olinadi

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4. Texnik Arxitektura

4.1 Umumiy Arxitektura Sxemasi

Tizim Arxitekturasi — High Level Overview

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                    FRONTEND (Web + Mobile)                                   │
│  Dashboard (React) │ Mobile App (React Native / PWA)                         │
└──────────────────────────────────────────────────────────────────────────────┘
                                │
                                │ HTTPS / WebSocket
                                │
┌──────────────────────────────────────────────────────────────────────────────┐
│                  API GATEWAY (nginx + SSL)                                   │
└──────┬──────────────────────┬──────────────────────┬──────────────────────┘
       │                      │                      │
       │                      │                      │
┌──────┴──────┐   ┌──────────┴──────────┐   ┌──────────┴──────────┐
│ Auth │   │ Main API │   │ Stream API │
│ Svc │   │ (FastAPI) │   │ (WebRTC)   │
└──────┬──────┘   └──────────┬──────────┘   └──────────┬──────────┘
       │                      │                      │
┌──────┴──────────────────────┴──────────────────────┴──────────────────────┐
│                  MESSAGE BROKER (Redis + Kafka)                              │
└──────┬──────────────────────┬──────────────────────────────────────────────┘
       │                      │
       │                      │
┌──────┴──────┐   ┌──────────┴──────────────────────────┐
│ Alert│   │   AI PROCESSING ENGINE   │
│ Svc │   │ (Python + YOLO + FaceNet) │
└──────┬──────┘   └──────────┬──────────────────────────┘
       │                      │
       │                      │
┌──────┴──────┐   ┌──────────┴──────────┐   ┌──────────────────┐
│ FCM │   │ CAMERA    │   │ PostgreSQL │
│ SMS │   │ RTSP FEEDS │   │ + Redis    │
└──────────┘   └──────────────────┘   └──────────────────┘
```

4.2 Technology Stack

4.2.1 Backend

KOMPONENT	TEXNOLOGIYA
API Framework	FastAPI (Python 3.11+) — async, yuqori samaradorlik
AI/ML Engine	PyTorch + ONNX Runtime — GPU/CPU inference
Object Detection	YOLOv8 (Ultralytics) — real-vaqt obyekt aniqlash

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Face Recognition	InsightFace (ArcFace backbone) — yuz tanish
Pose Estimation	MediaPipe Pose — yiqilish aniqlash
Action Recognition	SlowFast Networks — harakatni tahlil qilish
Video Processing	FFmpeg + OpenCV — stream va frame processing
Message Broker	Apache Kafka — high-throughput event streaming
Cache	Redis — session, cache, real-vaqt state
Primary Database	PostgreSQL 15+ — asosiy ma'lumotlar bazasi
Object Storage	MinIO (S3-compatible) — video va rasm saqlash
Authentication	JWT + OAuth2 (Keycloak) — xavfsiz autentifikatsiya
SMS Gateway	Playmobile / Eskiz.uz — O'zbekiston SMS
Push Notification	Firebase Cloud Messaging (FCM)
WebRTC Server	mediasoup — past kechikishli video stream

4.2.2 Frontend

KOMPONENT	TEXNOLOGIYA
Web Framework	Next.js 14 (React) — SSR, SEO, performance
UI Library	shadcn/ui + Tailwind CSS — zamonaviy dizayn
State Management	Zustand + React Query — server state
Real-time	Socket.io Client — WebSocket notifications
Video Player	Video.js + WebRTC — kamera streamlarini ko'rsatish
Charts/Analytics	Recharts — statistika va hisobotlar
Forms	React Hook Form + Zod — form validatsiya
Mobile	Progressive Web App (PWA) — ilovani yuklab olmasdan

4.2.3 DevOps & Infrastructure

KOMPONENT	TEXNOLOGIYA
Containerization	Docker + Docker Compose (dev) / Kubernetes (prod)
CI/CD	GitHub Actions — avtomatik deploy pipeline
Reverse Proxy	nginx — SSL termination, load balancing
Monitoring	Prometheus + Grafana — tizim monitoringi
Logging	ELK Stack (Elasticsearch + Logstash + Kibana)
Server (MVP)	2x vCPU, 8GB RAM, GPU (NVIDIA T4) — on-premise yoki cloud

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5. Veb-sayt Talablari (MVP)

5.1 Sahifalar Ro'yxati

SAHIFA	URL	KIRISH RUXSATI
Kirish (Login)	/login	Barcha foydalanuvchilar
Asosiy Dashboard	/dashboard	Barcha rollar (rol asosida ko'rinish)
Kamera Lenta	/cameras	Barcha rollar (rol asosida filtr)
Kamera Detail	/cameras/:id	Barcha rollar
Hodisalar Logi	/events	Barcha rollar
Hodisa Detail	/events/:id	Barcha rollar
Bolalar Ro'yxati	/children	Tarbiyachi (o'z guruhi), Mudir (barchasi)
Bola Profili	/children/:id	Tarbiyachi, Mudir
Xodimlar	/staff	Faqat Mudir
Guruhlar	/groups	Faqat Mudir
Hisobotlar	/reports	Faqat Mudir
Sozlamalar	/settings	Faqat Mudir
Profil	/profile	Barcha rollar

5.2 Dashboard — Asosiy Ekran

Tarbiyachi Dashboardi

- Guruh bo'yicha real-vaqt kamera ko'rinishi (asosiy kamera full-width)
- Guruhda nechta bola mavjud (bugun kelgan/kelmagan)
- So'nggi hodisalar listi (faqat o'z guruhi)
- Alert panel — yangi bildirishnomalar

Xavfsizlik Xodimi Dashboardi

- Kirish/chiqish kameralari (tashqi perimetr) — multi-view grid
- So'nggi 24 soat ichidagi xavfsizlik hodisalari
- Hozir hududda bo'lgan bolalar soni
- Begona shaxslar ro'yxati (agar aniqlangan bo'lsa)

Mudir Dashboardi

- Overview kartochkalar: Jami bolalar, Bugun kelganlar, Faol hodisalar, Online kameralar

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- Barcha guruhlar holati — mini-karta ko'rinish
- Oxirgi hodisalar lenti (barcha guruhlar)
- Xodimlar faolligi (kim qachon kirgan/chiqqan)
- Tezkor statistika grafiklar

5.3 Kamera Ko'rish Sahifasi

- Grid layout: 1x1, 2x2, 3x3 ko'rinishlarni almashtirish
- Har bir kamera uchun: nomi, guruhi, holati (online/offline), so'nggi hodisa
- Kamera ustiga bosish → full-screen ko'rinish
- PTZ kameralar uchun boshqaruv (agar mavjud bo'lsa)
- Video arxiv — vaqt bo'yicha orqaga qaytish (playback)
- Snapshot olish va yuklab olish imkoniyati

5.4 Hodisalar Logi Sahifasi

- Filtrlar: Sana, Guruh, Hodisa turi, Daraja (Kritik/Ogohlantirish/Axborot)
- Har bir hodisa uchun: vaqt, guruh, bola ismi, hodisa turi, javob vaqti
- Hodisaga tegishli video klipni ko'rish (avtomatik saqlangan 30 soniyalik klip)
- Hodisani 'Hal etildi' deb belgilash imkoniyati
- Eksport: PDF yoki Excel formatida hisobot chiqarish

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6. API Dizayni

6.1 REST API Endpoints

API Base URL
Production: https://api.bogchaGuard.uz/v1
Development: http://localhost:8000/v1
Autentifikatsiya: Bearer Token (JWT) — barcha endpointlarda majburiy

6.1.1 Autentifikatsiya

ENDPOINT	TAVSIF
POST /auth/login	Tizimga kirish — token olish
POST /auth/refresh	Access tokenni yangilash
POST /auth/logout	Tizimdan chiqish — tokenni bekor qilish
GET /auth/me	Joriy foydalanuvchi ma'lumotlari

6.1.2 Kameralar

ENDPOINT	TAVSIF
GET /cameras	Barcha kameralar ro'yxati (rol asosida filtr)
GET /cameras/:id	Bir kamera ma'lumotlari
GET /cameras/:id/stream	RTSP stream URL olish
GET /cameras/:id/snapshot	So'nggi frame (JPEG)
POST /cameras	Yangi kamera qo'shish (faqat Mudir)
PUT /cameras/:id	Kamera ma'lumotlarini yangilash
DELETE /cameras/:id	Kamerani o'chirish

6.1.3 Hodisalar (Events)

ENDPOINT	TAVSIF
GET /events	Hodisalar ro'yxati (filtr: date, group, type, severity)
GET /events/:id	Hodisa detallari + video klip URL
PUT /events/:id/resolve	Hodisani hal etildi deb belgilash
GET /events/stats	Statistika (count by type, by group, by day)
GET /events/:id/video	Hodisaga tegishli video klipni stream qilish

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6.1.4 Bolalar va Xodimlar

ENDPOINT	TAVSIF
GET /children	Bolalar ro'yxati (tarbiyachi: o'z guruhi)
POST /children	Yangi bola qo'shish + yuz rasmi yuklash
GET /children/:id	Bola profili + hodisalar tarixi
PUT /children/:id	Bola ma'lumotlarini yangilash
POST /children/:id/face	Yuz rasmi yangilash (biometrik)
GET /staff	Xodimlar ro'yxati (faqat Mudir)
POST /staff	Yangi xodim qo'shish
PUT /staff/:id	Xodim ma'lumotlarini yangilash

6.2 WebSocket Events

Real-vaqt xabardorliklar uchun WebSocket (Socket.io) ishlatiladi:

EVENT NOMI	MA'LUMOT
alert:new	Yangi hodisa — daraja, turi, guruh, bola ismi, vaqt
alert:resolved	Hodisa hal etildi
camera:offline	Kamera ulanishi uzildi
camera:online	Kamera qayta ulandi
child:entered	Bola bog'chaga kirdi (face recognition)
child:exited	Bola bog'chadan chiqdi
intruder:detected	Noma'lum shaxs aniqlandi — koordinatlar

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7. Ma'lumotlar Modeli (Database Schema)

7.1 Asosiy Jadvallar

users — Foydalanuvchilar

```
id UUID PRIMARY KEY
email VARCHAR(255) UNIQUE NOT NULL
password_hash VARCHAR(255) NOT NULL
full_name VARCHAR(255) NOT NULL
role ENUM('tarbiyachi', 'security', 'mudir') NOT NULL
group_id UUID REFERENCES groups(id)   -- tarbiyachi uchun
face_embedding VECTOR(512)            -- biometrik
is_active BOOLEAN DEFAULT true
created_at TIMESTAMP DEFAULT NOW()
last_login TIMESTAMP
```

children — Bolalar

```
id UUID PRIMARY KEY
full_name VARCHAR(255) NOT NULL
date_of_birth DATE NOT NULL
group_id UUID REFERENCES groups(id) NOT NULL
parent_phone VARCHAR(20)
parent_name VARCHAR(255)
medical_notes TEXT
face_embeddings VECTOR(512)[]         -- multiple photos
photo_url VARCHAR(500)
is_active BOOLEAN DEFAULT true
created_at TIMESTAMP DEFAULT NOW()
```

cameras — Kameralar

```
id UUID PRIMARY KEY
name VARCHAR(255) NOT NULL
location VARCHAR(255)                 -- xona/joylashuv nomi
rtsp_url VARCHAR(500) NOT NULL
group_id UUID REFERENCES groups(id)   -- NULL = tashqi/umumiy
camera_type ENUM('indoor', 'outdoor', 'entrance', 'perimeter')
is_ptz BOOLEAN DEFAULT false
status ENUM('online', 'offline', 'error') DEFAULT 'offline'
last_seen TIMESTAMP
created_at TIMESTAMP DEFAULT NOW()
```

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events — Hodisalar

```
id UUID PRIMARY KEY
event_type VARCHAR(100) NOT NULL      -- fall, fight, intruder, exit...
severity ENUM('critical', 'warning', 'info') NOT NULL
camera_id UUID REFERENCES cameras(id)
child_id UUID REFERENCES children(id) -- NULL if unknown
group_id UUID REFERENCES groups(id)
description TEXT
video_clip_url VARCHAR(500)           -- 30 soniyalik klip
thumbnail_url VARCHAR(500)
ai_confidence FLOAT                   -- 0.0 - 1.0
metadata JSONB                        -- qo'shimcha ma'lumotlar
status ENUM('active', 'resolved', 'false_positive') DEFAULT 'active'
resolved_by UUID REFERENCES users(id)
resolved_at TIMESTAMP
created_at TIMESTAMP DEFAULT NOW()
```

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8. Xavfsizlik Talablari

8.1 Ma'lumot Xavfsizligi

- Barcha ma'lumotlar AES-256 shifrlash bilan saqlash
- Transport xavfsizligi: TLS 1.3 (HTTPS, WSS)
- Biometrik ma'lumotlar (yuz rasmi embeddinglari) alohida xavfsiz bazada
- Foydalanuvchi parollari bcrypt (cost factor 12) bilan hash qilish
- Video arxivlar shifrlangan storage'da saqlash
- GDPR va O'zbekiston 'Shaxsiy ma'lumotlar to'g'risida'gi qonunga muvofiqlik

8.2 Kirish Nazorati

- JWT token umri: Access Token — 15 daqiqa, Refresh Token — 7 kun
- RBAC (Role-Based Access Control) — har bir API endpoint rol asosida himoya
- Brute-force himoya: 5 ta noto'g'ri urinishdan so'ng 15 daqiqa blok
- Audit log: barcha muhim amallar (login, data o'zgarish) yozib olinadi
- IP whitelisting — admin panelga faqat ruxsat etilgan IP manzillardan kirish

8.3 Maxfiylik

- Video ma'lumotlarga faqat vakolatli xodimlar kirishi mumkin
- Bolalar biometrik ma'lumotlarini 3rd party bilan ulashish qat'iyan ta'qiqlangan
- Arxiv videolar 30 kundan keyin avtomatik o'chiriladi
- Ota-onalar ma'lumotlarini eksport qilish va o'chirish imkoniyati (GDPR-like)

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9. Ishlab Chiqish Bosqichlari

9.1 MVP (Phase 1) — 3 Oy

MVP Doirasi

Sprint 1-2 (Hafta 1-4): Arxitektura, DB dizayn, Auth service, Camera RTSP ingestor
Sprint 3-4 (Hafta 5-8): YOLOv8 integratsiya, Fall/Fight detection, Alert service
Sprint 5-6 (Hafta 9-12): Web dashboard (React), Role-based UI, WebSocket notification

MVP deliverables:

- 2-3 ta kamera bilan real-vaqt monitoring
- Fall va Fight detection ishlaydi
- 3 roldagi foydalanuvchilar tizimi
- Web dashboard (responsive)
- Push notification (in-app)

9.2 Phase 2 — 2 Oy

- Face Recognition — bolalar va xodimlarni avtomatik tanish
- Intruder Detection — noma'lum shaxslarni aniqlash
- SMS va Email notification integratsiya
- Video arxiv va playback funksiyasi
- Mobilga moslashtirish (PWA)
- Hisobotlar moduli (PDF eksport)

9.3 Phase 3 — 2 Oy

- Audio tahlil — yig'lash, qichqirish aniqlash
- Hudud monitoring (Zone Analytics) — xavfli zonalar belgilash
- Ota-onalar uchun alohida portal
- Ko'p bog'chani boshqarish (multi-tenant)
- AI modellarini fine-tuning — mahalliy ma'lumotlar asosida

9.4 Team Composition

LAVOZIM	SONI	ASOSIY VAZIFALAR
Senior Backend Engineer	2	FastAPI, PostgreSQL, Kafka, Auth

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AI/ML Engineer	2	YOLO, FaceNet, SlowFast, Pipeline
Frontend Engineer	2	React, Next.js, Dashboard, UI/UX
DevOps Engineer	1	Docker, CI/CD, nginx, monitoring
QA Engineer	1	Test strategy, automation, security testing
Product Manager	1	Talablar, stakeholder, sprint planning
UI/UX Designer	1	Dizayn tizimi, prototip, user testing

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10. Funksional Bo'lmagan Talablar

10.1 Samaradorlik (Performance)

KO'RSATKICH	MAQSAD
AI inference kechikishi	< 200ms (per frame, single GPU)
Alert yuborish vaqti	< 3 soniya (hodisa aniqlanganidan)
API javob vaqti	< 200ms (p95 percentile)
Video stream kechikishi	< 2 soniya (WebRTC)
Dashboard yuklanish	< 1.5 soniya (LCP)
Parallel kameralar	20+ kamera (bitta server, GPU bilan)
Concurrent foydalanuvchilar	100+ simultaneous users

10.2 Ishonchlilik (Reliability)

- Uptime SLA: 99.5% (oyiga < 3.6 soat to'xtash)
- Kamera uzilsa — avtomatik qayta ulanish (30 soniyada 1 marta urinish)
- Database backup: har kuni avtomatik, 30 kun saqlanadi
- Failover: asosiy server ishlamasa — 5 daqiqa ichida backup server

10.3 Kengaytirilishi (Scalability)

- Horizontal scaling: yangi AI processing node qo'shish — zero downtime
- Multi-tenant: bir tizimda bir nechta bog'cha (Phase 3)
- Kamera soni: 5 dan 100+ ga qadar kengaytirish imkoniyati

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11. Risklar va Yechimlar

RISK	EHTIMOLLIK	YECHIM
AI modeli ko'p yolg'on pozitiv berishi	O'rtacha	Confidence threshold sozlash, human-in-loop verification
GPU server qimmatligi	Yuqori	MVP uchun CPU inference, keyinchalik GPU cloud
RTSP kamera muvofiqlik muammosi	O'rtacha	ONVIF standard, FFmpeg adapter layer
Biometrik ma'lumotlar xavfsizligi	Past	Isolated DB, encryption, audit logs
Internet uzilganda ishlash	O'rtacha	Edge computing — local AI inference, offline mode
Yolg'on alarm — panic	Yuqori	Daraja tizimi, human confirmation for critical alerts
Ma'lumot hajmi (video storage)	Yuqori	H.264 compression, 30 kunlik retention policy

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12. Qabul Qilish Mezonlari (MVP)

12.1 Funksional Testlar

- Tarbiyachi o'z guruhi kamerasini ko'ra oladi — boshqa guruhnikini emas ✓
- Fall Detection — 90%+ aniqlik (test video to'plami asosida) ✓
- Alert 3 soniya ichida dashboardda ko'rinadi ✓
- Mudir barcha guruhlarni ko'ra oladi ✓
- Xavfsizlik xodimi tashqi kameralargina ko'ra oladi ✓
- Tizimga kirish autentifikatsiyasiz imkonsiz ✓

12.2 Samaradorlik Testlar

- 3 ta parallel kamera stream — CPU < 80%, RAM < 6GB ✓
- Dashboard 100 foydalanuvchida sekinlamasdan ishlaydi ✓
- API stress test: 500 req/sec — error rate < 1% ✓

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Ilova: Glossariy

ATAMA	TA'RIFI
Computer Vision	Kompyuterning rasm va videolarni tahlil qilish qobiliyati
RTSP	Real Time Streaming Protocol — kamera videosini uzatish protokoli
YOLO	You Only Look Once — real-vaqt obyekt aniqlash algoritmi
FaceNet/ArcFace	Yuzni tanish uchun ishlatiladigan neyron tarmoq modeli
Pose Estimation	Odamning gavda holatini aniqlash texnologiyasi
RBAC	Role-Based Access Control — rol asosida kirish nazorati
JWT	JSON Web Token — xavfsiz autentifikatsiya uchun token formati
WebRTC	Web Real-Time Communication — brauzerda real-vaqt video
WebSocket	Real-vaqt ikki tomonlama muloqot protokoli
Kafka	Yuqori samarali message broker (event streaming platform)
MinIO	S3-compatible open-source object storage
PWA	Progressive Web App — web-sayt sifatida ishlovchi mobil ilova
Inference	AI modelining bashorat qilish jarayoni
Edge Computing	Ma'lumotni markaziy serverga yubormay, mahalliy qurilmada qayta ishlash
ONVIF	Open Network Video Interface Forum — kamera standartlari

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